

# Linear models

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Jeff Leek

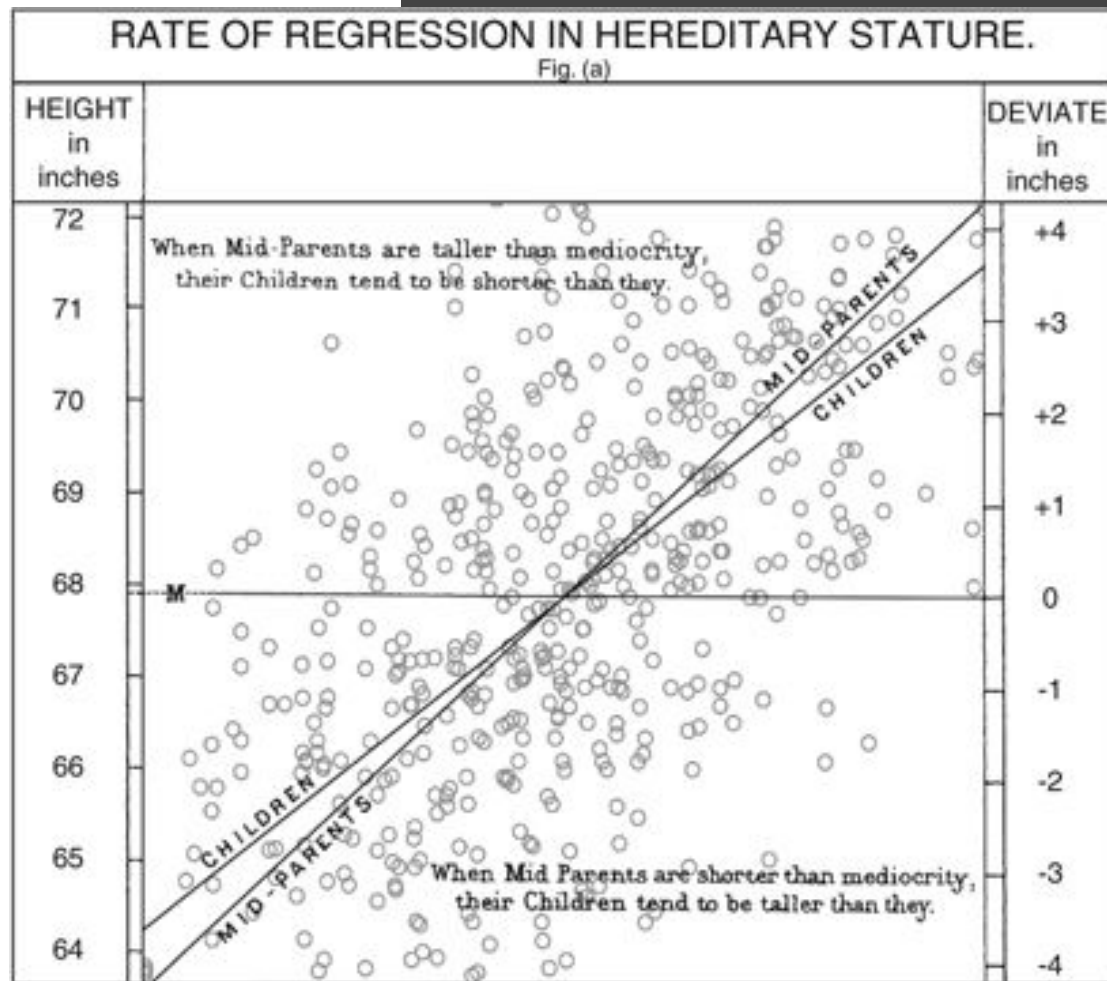
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# Basic idea

- Fit the “best line” relating two variables
- In math we are minimizing the relationship  $(Y - b_0 - b_1X)^2$
- You can always fit a line, the question is whether it is a good fit or not

**An old, but really useful idea!**



**Still relevant everywhere in  
genomics**

## Article

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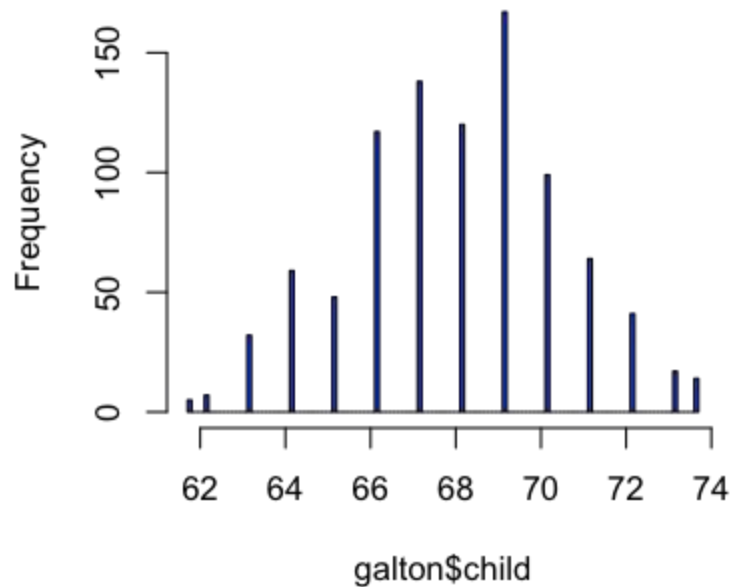
*European Journal of Human Genetics* (2009) **17**, 1070–1075; doi:10.1038/ejhg.2009.5; published online 18 February 2009

# Predicting human height by Victorian and genomic methods

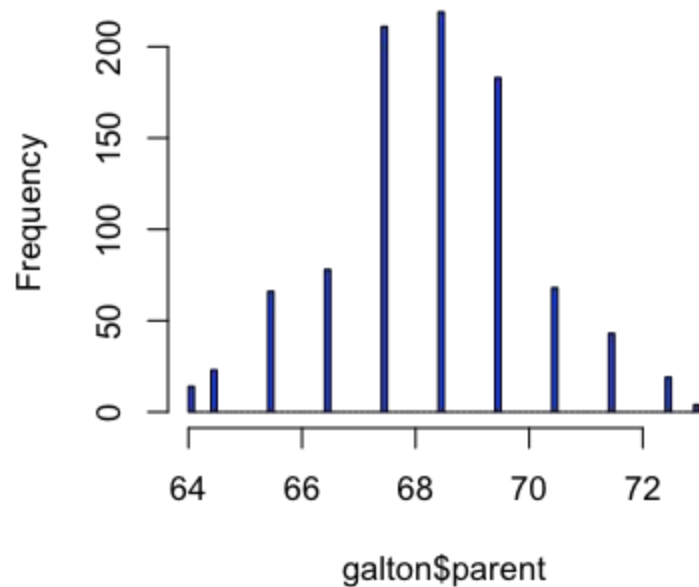
Yurii S Aulchenko<sup>1,2,7</sup>, Maksim V Struchalin<sup>1,3,7</sup>, Nadezhda M Belonogova<sup>2,4</sup>, Tatiana I Axenovich<sup>2</sup>, Michael N Weedon<sup>5</sup>, Albert Hofman<sup>1</sup>, Andre G Uitterlinden<sup>6</sup>, Manfred Kayser<sup>3</sup>, Ben A Oostra<sup>1</sup>, Cornelia M van Duijn<sup>1</sup>, A Cecile J W Janssens<sup>1</sup> and Pavel M Borodin<sup>2,4</sup>

# The Galton example explained

**Histogram of galton\$child**

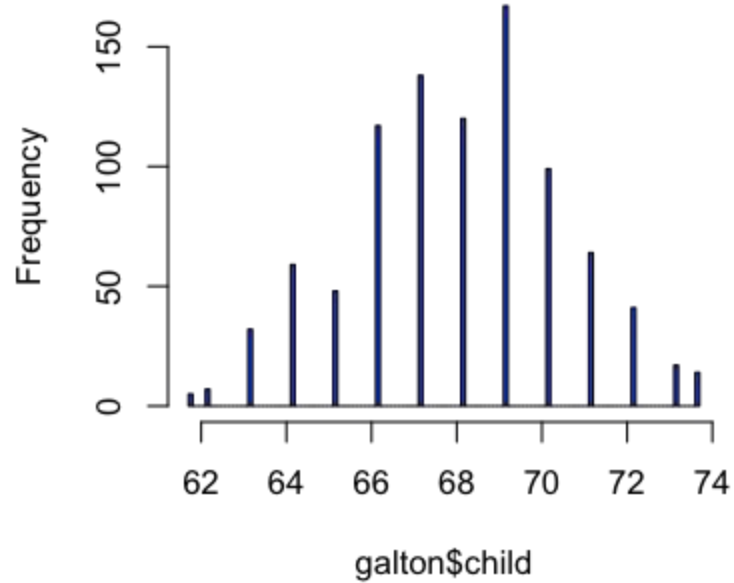


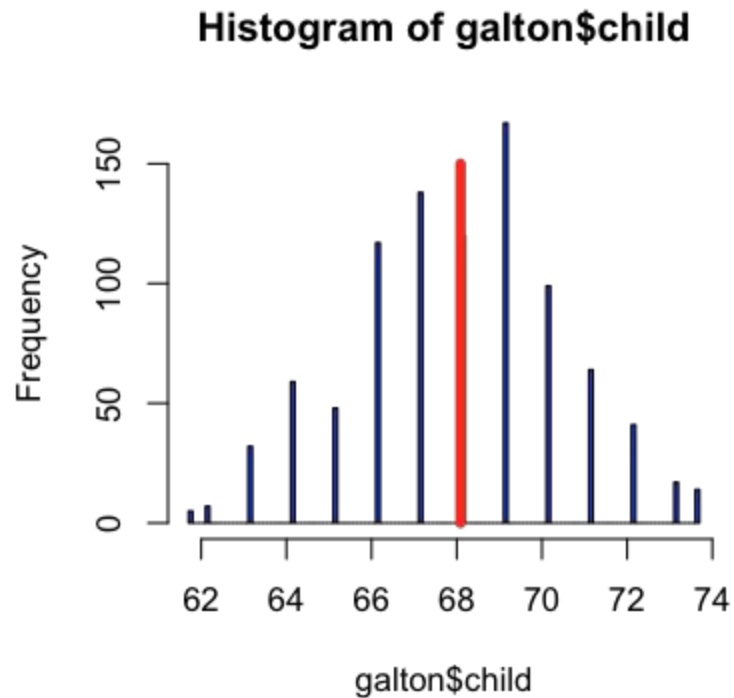
**Histogram of galton\$parent**



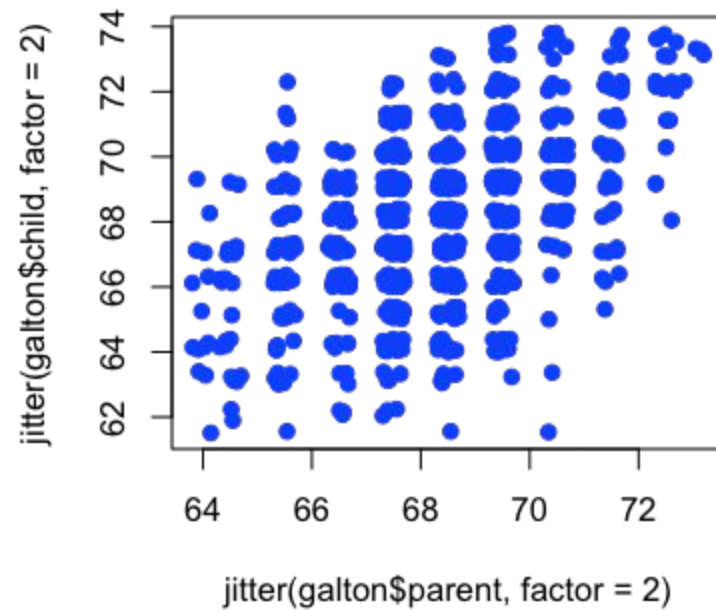


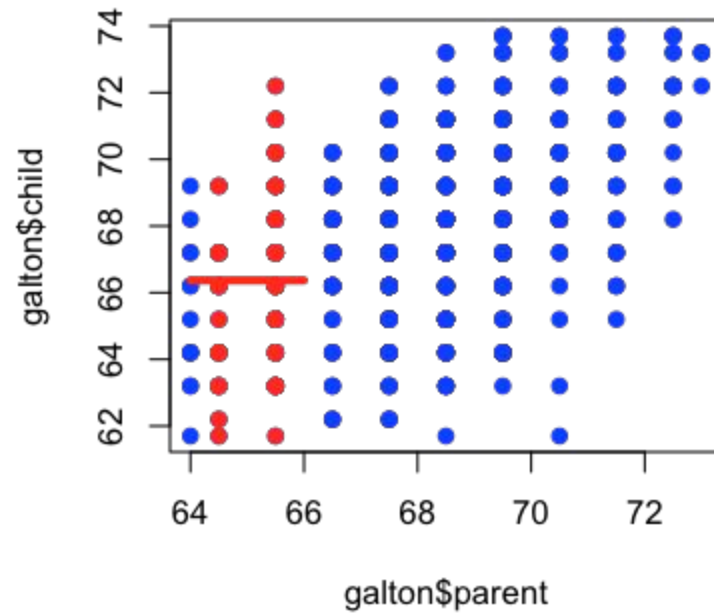
**Histogram of galton\$child**

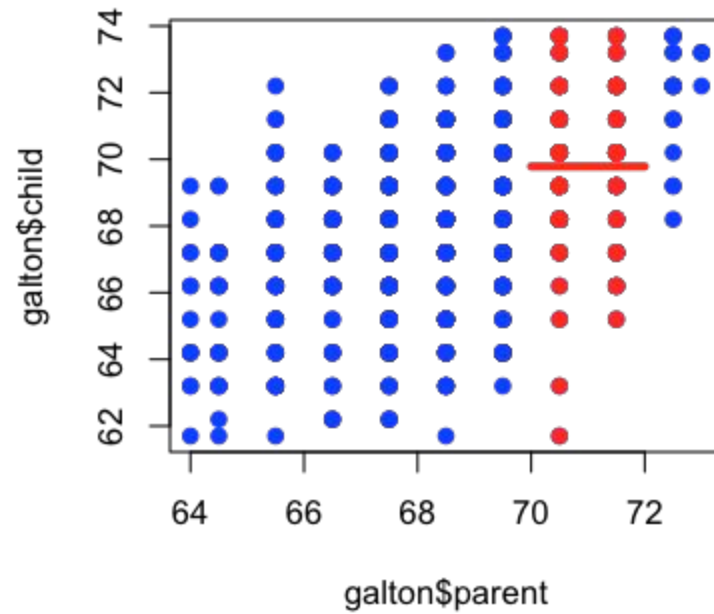


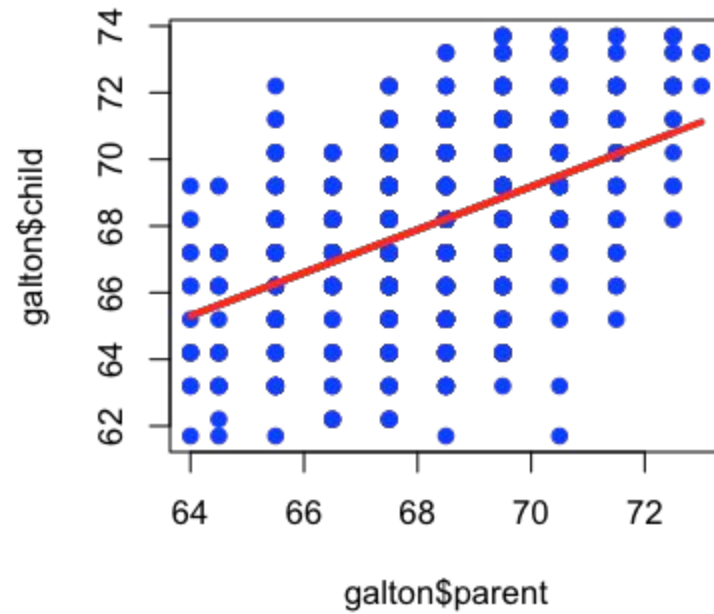


Minimizes  $\sum(Y - c)^2$



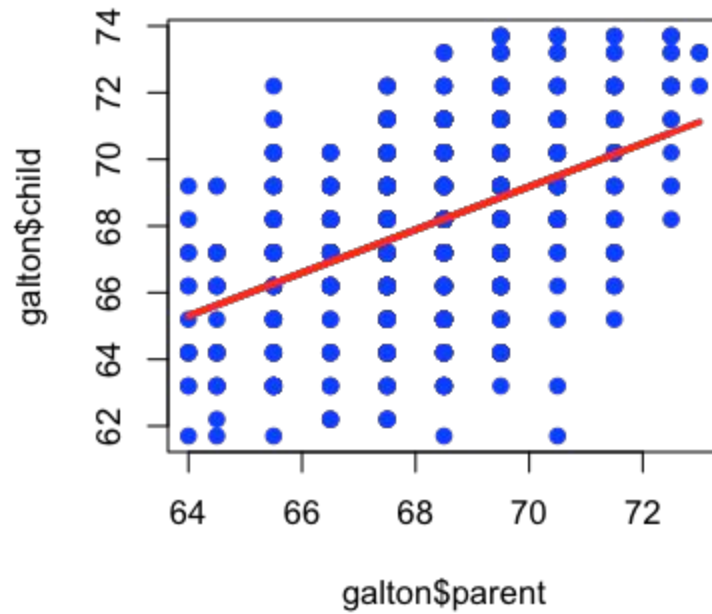






Equation for a line:

$$C = b_0 + b_1 P$$





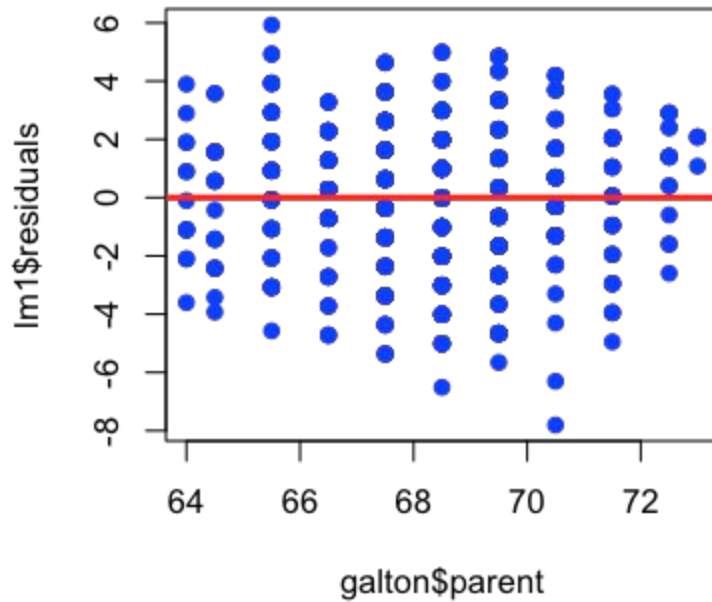
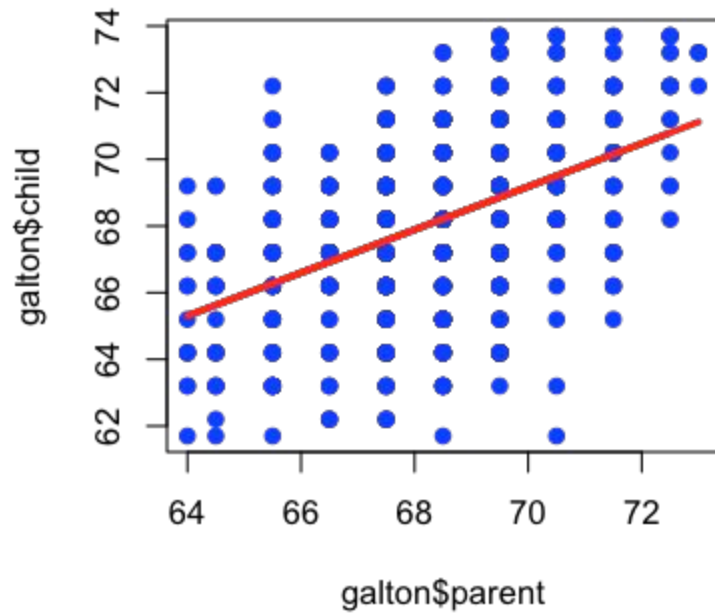
# Equation with noise

$$C = b_0 + b_1 P + e$$

Fit by minimizing

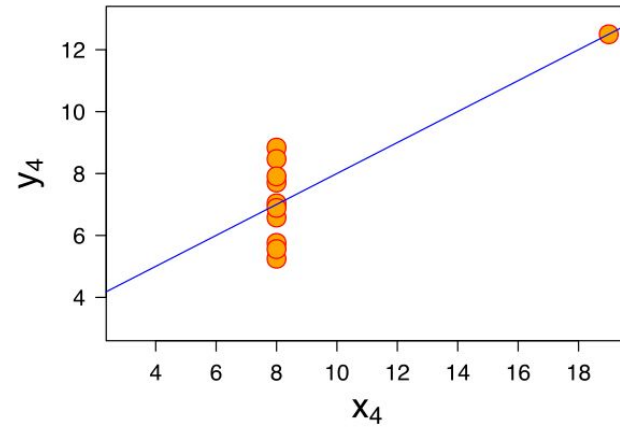
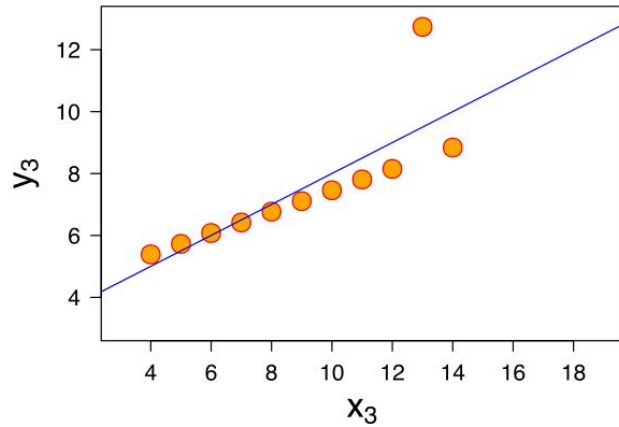
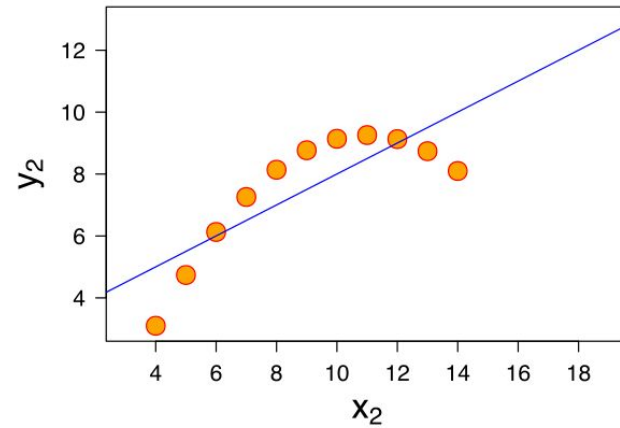
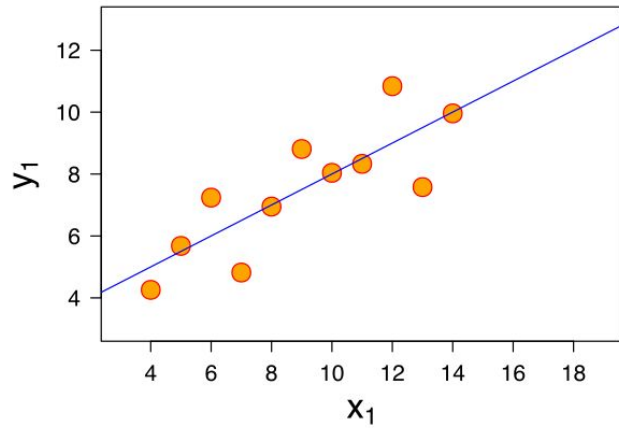
$$\sum (C - b_0 - b_1 P)^2$$

**Check residuals**



You can always fit a line

But it might not be the right thing



# Notes and further reading

- Linear models is a whole class (no joke):

<https://www.coursera.org/course/regmods>

- Basic thing to keep in mind is does the line fit?
- Great additional notes in Chapter 2 here:

<http://genomicsclass.github.io/book/>